

Optimization for Machine Learning

Exercise sheet **Influence of decompositions to algorithm efficiency**

Bogdan Savchynskyy - Bogdan.Savchynskyy@tu-dresden.de

Alexander Kirillov - Alexander.Kirillov@tu-dresden.de

The task of this exercise is to compare how the decomposition selection influences performance of optimization methods. We consider the **EDGE** and **GRID** decompositions and use both with the TRW-S algorithm. The **EDGE** decomposition splits the master graph into one-edge subgraphs (the edge and its two incident nodes), i.e. there is one subproblem corresponding to each edge of the initial graph. The **GRID** decomposition splits the input grid graph into separate columns and rows.

Both decompositions are acyclic, therefore both correspond to the same optimal dual value. However, the speed of algorithm convergence may depend on the decomposition. In particular, the latter influences the fact, how fast the information is propagated through the graph. We will consider 3 datasets:

1. **Random**, with unary and pairwise factors generated randomly and independently. In this type of models, the optimal label in one part of the graph is almost independent on the optimal label in another part of the graph. Therefore, no information propagation is required.
2. **Color segmentation** dataset. In this dataset the weight of pairwise factors (Potts model) is significantly lower than those of unary factors. Therefore, the dependence between graph nodes decreases fast with the distance between them. Minor information propagation is required.
3. **Stereo** dataset. Unary potentials of this dataset often contain many similar values, the optimal one can be selected only with the help of pairwise factors. Changes in labelings of one node may propagate along a big portion of the graph.

The experiments will be performed with a variant TRW-Si (internal OpenGM implementation) of the TRW-S algorithm.

Exercise 1 (Random dataset). Run TRW-Si algorithm on the random dataset with **EDGE** decomposition:

```
opengm_min_sum -a TRWSi -p 1 -v --maxIt 300 -d EDGE -o synthetic-edge.h5
-m synthetic200x100x5_amp1.0_s1001.0.h5
```

with **GRID** decomposition:

```
opengm_min_sum -a TRWSi -p 1 -v --maxIt 300 -d GRID -o synthetic-grid.h5
-m synthetic200x100x5_amp1.0_s1001.0.h5
```

Build the plot, how primal (energy) and dual objectives evolve in time:

```
python print-plot.py -c synthetic-edge.h5 synthetic-grid.h5
```

Exercise 2 (Color segmentation). Repeat the experiments from Exercise 1 with several problem instances from the `color-seg-n4` (download at <http://hci.iwr.uni-heidelberg.de/opengm2/> or use the downloaded ones from the previous exercises).

Exercise 3 (Stereo). Repeat the experiments from Exercise 1 with the `tsu-gm` and `ven-gm` datasets (download at <http://hci.iwr.uni-heidelberg.de/opengm2/> or use the downloaded ones from the previous exercises).

As a result of the exercise make a document file (in pdf, odt, odp, doc, docx, ppt or pptx format) with the obtained plots and your conclusions.